



TECHNISCHE  
UNIVERSITÄT  
DARMSTADT

# Simulation Studies on Active Car Spoiler

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**Submitted by Sritej Sritej & Akhil Unikrishnan as a part of "Tutorium Numerical Simulation of Flow problems."**

**Master's in Aerospace Engineering**

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## **1 Contents**

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|          |                                                                                               |
|----------|-----------------------------------------------------------------------------------------------|
| 1        | Introduction                                                                                  |
| 2.1      | Active Car Spoiler                                                                            |
| 2.2      | Mission Statement                                                                             |
| 2.3      | Project Goals                                                                                 |
| 3        | Theoretical Background                                                                        |
| 4        | CAD Model                                                                                     |
| 4.1      | Modelling Methodology                                                                         |
| 4.2      | Flow Domain, Geometry and Assembly Preparation, Mesh Generation and Scence and Representation |
| 4.3      | Boundary Conditions                                                                           |
| 4.4      | Physics Models                                                                                |
| 5        | Results and Discussion                                                                        |
| 5.1      | Mesh Convergence Study                                                                        |
| 5.2      | Mesh Quality Check and Results Validation                                                     |
| 6        | Conclusion                                                                                    |
| 7        | References                                                                                    |
| Appendix |                                                                                               |

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## 2 Introduction

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### 2.1 Active Car Spoiler

An active car spoiler possesses the capability to adjust its angle of attack, thereby enabling the achievement of various levels of downforce and drag in response to different driving scenarios. Consequently, the angle at which the spoiler rests on the car will dynamically change according to real-time driving conditions. It is important to note that the vertical and lateral positioning of the spoiler remains fixed while the car is in motion, with adjustments solely made to the angle of the spoiler.

### 2.2 Mission Statement

In the burgeoning realm of Formula 1 racing platforms, conducting analyses of various factors proves instrumental in comprehending the intricacies of a Formula 1 car, particularly in relation to key racing variables such as wind velocity, coefficient of lift, drag, and Angle of Attack. Employing computational flow analysis of the car spoiler system enables manufacturers to discern the optimal range of angle of attacks. This insight not only empowers drivers to exercise better control over their vehicles but also facilitates the maximization of a racing car's potential to secure victories while mitigating the risks of accidents.

### 2.3 Project Goals

Our primary objective is to ascertain the Angle of Attack (AOA) that yields maximum downforce and drag forces across a range of speeds. Additionally, we aim to delineate the angle range conducive to optimizing drag and downforce values. In practical driving scenarios, these findings prove invaluable during braking, acceleration, and cornering maneuvers.

The outcomes of our endeavor can serve as inputs for an electronic feedback system, enabling real-time adjustment of the spoiler's airflow angle. Our project endeavors to dynamically adapt the angle of attack to engender optimal downforce and drag scenarios reflective of the car's driving dynamics. During cornering, for instance, maximizing downforce becomes paramount, necessitating varying degrees of AOA depending on speed and turning angle. Conversely, during linear acceleration, minimizing drag is essential for smooth motion. Similarly, when decelerating rapidly, deploying maximum drag via the spoiler aids in swift braking.

Realizing this in practice necessitates the spoiler's ability to dynamically adjust its AOA using an actuator. Additionally, it must discern the prevailing driving conditions using data from accelerometers and speedometers. This process mirrors the manipulation of the Spoiler domain within STAR-CCM+ through the Transform command, wherein a variable such as AOA governs the rotation of the domain to reflect changing driving conditions

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## 3 Theoretical Background

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Airfoils and wings exhibit varying force levels contingent upon the angle of attack, with a critical angle marking a significant decrease in forces due to airflow separation. This critical angle varies among airfoil designs tailored to diverse aerodynamic scenarios.

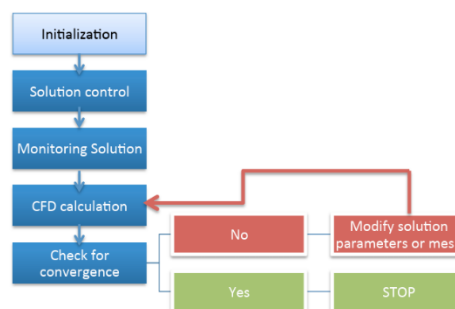


Figure 1 : Steps in CFD

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In the research conducted by Divya Teja Ayyagari and Yuping He, the analysis of downforce generated by an adjustable spoiler was conducted using Computational Fluid Dynamics (CFD) and wind tunnel experiments. While akin to our data collection objectives, the study primarily focused on maximizing the downforce-to-drag ratio. It did not account for instances where drag might be advantageous during braking, such as for aerobraking purposes. Furthermore, the study primarily investigated high-speed scenarios, whereas we aim to extend our analysis across a broader spectrum of speeds and driving conditions.

Nonetheless, the study elucidates that increasing the angle of attack augments both downforce and drag until a critical angle is reached, beyond which flow separation diminishes downforce benefits and compromises the efficacy of the spoiler. Moreover, the study by Jingdong Cai, Tushita Sikder, Saurabh Kapoor, and Yuping He underscores the potential of active spoilers in enhancing driving dynamics, although further details regarding these The performance of active spoilers in emergency situations, particularly during tight cornering maneuvers at very high speeds, has been a focal point of recent research. Studies have indicated that active spoilers enable vehicles to withstand higher lateral accelerations typical of high-speed cornering maneuvers. The active spoiler examined in these studies has demonstrated four key improvements in driving performance: enhanced tire capability to generate larger cornering forces, stabilization of the vehicle at high speeds, improved braking performance, and increased traction.

While the insights from these studies serve as valuable benchmarks, our project aims to expand upon them by examining a wider range of speeds and incorporating varied criteria for the downforce-to-drag ratio based on the current driving conditions of the car. These articles provided essential groundwork for our project, shedding light on the significant impact different angles of attack can have on driving performance and inspiring us to develop a product with greater depth.

The additional research and development we have undertaken are aimed at creating a spoiler that surpasses existing ones in performance while remaining more cost-effective. Through this enhanced level of innovation, we strive to produce a spoiler that not only outperforms its counterparts but also offers superior affordability, thereby advancing the standards of automotive aerodynamics.

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#### 4 CAD Model

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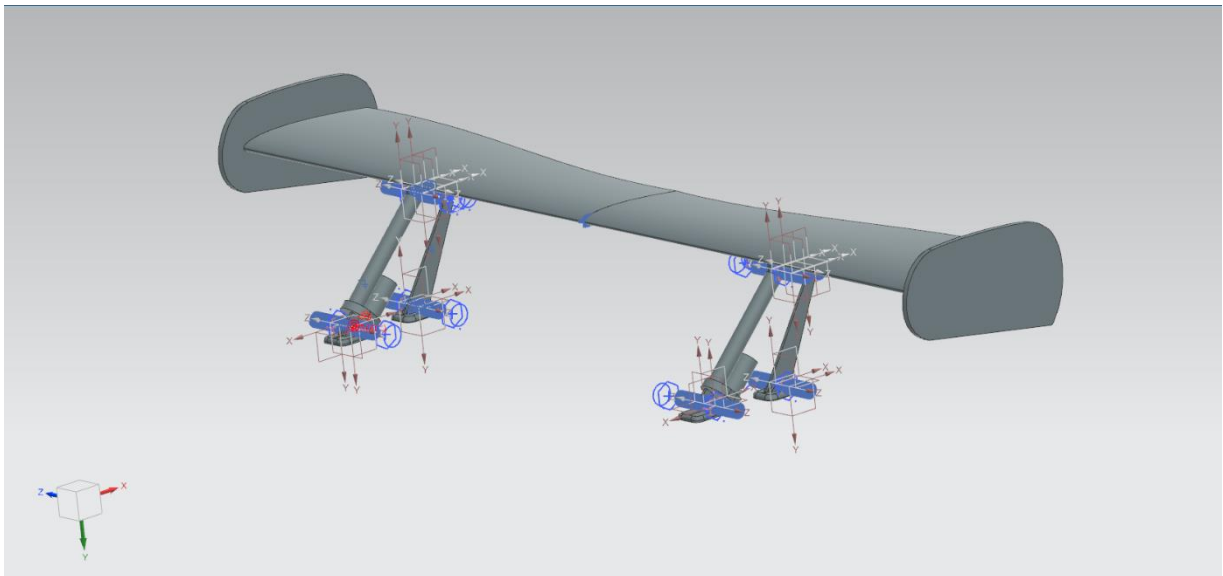


Figure 2: 3D CAD Assembly

After careful selection of the components, a CAD model was created using Siemens NX. Detailed drawings are included in Appendix. The CAD render pictured in Figure 1 includes two linear electronic actuators, two front mounting brackets, and a rear wing.

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#### 4.1 Modeling Methodology

1. **Conceptualization and Design Ideation:** The genesis of the modelling process resides in conceptualization and design ideation. Through collaborative brainstorming sessions and in-depth analysis of project requirements, the foundational principles and design objectives are delineated. This stage sets the trajectory for subsequent phases, providing a holistic framework for the realization of the CAD model.
2. **Preliminary Sketching and Layout Planning:** With the design concept elucidated, preliminary sketching and layout planning ensue. Utilizing traditional sketching tools or digital platforms, initial renderings and sketches are crafted to visualize key components and geometric configurations. This phase serves as a preliminary exploration of design aesthetics and functional requirements, facilitating iterative refinement and enhancement of design elements.
3. **CAD Modelling in Siemens NX:** The crux of the methodology resides in the CAD modelling phase, wherein the conceptualized design is translated into a tangible digital model using Siemens NX. Leveraging an array of powerful tools and functionalities embedded within the software suite, the modelling process is characterized by meticulous attention to detail and adherence to industry best practices.
4. **Geometry Creation and Parametric Modelling:** Geometry creation forms the cornerstone of the modelling process, wherein fundamental shapes and components are synthesized to form the skeletal structure of the model. Through the adept utilization of parametric modelling techniques, key design parameters are encapsulated within the model, affording flexibility and scalability in design iterations.
5. **Feature-based Design and Assembly Integration:** Feature-based design augments the modelling process, enabling the integration of intricate design elements and geometric features with unparalleled precision. From fillets and chamfers to complex surface blends, each feature is meticulously crafted to adhere to design specifications and aesthetic considerations. Furthermore, assembly integration facilitates the seamless incorporation of individual components, culminating in a comprehensive and cohesive CAD model.
6. **Iterative Refinement and Validation:** Iterative refinement serves as a hallmark of the modelling methodology, wherein the CAD model undergoes rigorous scrutiny and validation to ensure adherence to design specifications and functional requirements. Through iterative prototyping and simulation analyses, potential design flaws and inefficiencies are identified and rectified, fostering a culture of continuous improvement and optimization.

#### 4.2 Flow Domain, Geometry and Assembly Preparation, Mesh Formation and Scene Representations

The NACA0012-based spoiler was constructed using Siemens NX, leveraging its features such as Mirroring, Trimming, Sketching, and Constraint tools extensively. The basic geometry employed Studio Surface with G0 tangent curvature to generate effective surface cross-sections at equidistance intervals of 3 inches, along with scaling of the root profile. Components like the Back Mounting Plate, Powered Back Leg, Front Mounting Plate, and Front Leg were created using simple extrude and sketch commands, while the Powered Back Leg was assembled with extension supports.

Standard ISO values for Bolt and Nut were obtained from CADfiles.org to expedite modeling of essential components within the limited project timeframe. NX facilitated assembly and drafting processes seamlessly, utilizing features like "Make Work Part only" and "Open in Window" for quick edits and ensuring continuous geometry.

The PC specifications enabled parallel processing, employing a 4-core parallel processing setup to import parts with the "Import Surface options" configured to create one part surface per part. Due to the limited scope of aerodynamic forces analysis and to maintain relevance to basic aerodynamic theory, certain components like the Slide Plates were excluded from CFD flow analysis, focusing solely on the wing surface.

| PC Pool   |                                            |
|-----------|--------------------------------------------|
| Processor | 12 <sup>th</sup> Gen Intel Core i7-12700KF |
| Frequency | 3.60Hz                                     |
| RAM       | 32GB                                       |
| Cores     | 12                                         |
| Threads   | 20                                         |

Table 1: PC Specifications

A block of specific dimensions was constructed, and the Boolean Subtract command was utilized to obtain the Fluid Domain by subtracting the block from the wing. Surface generation involved splitting by the Patch command, followed by surface renaming and assignment of parts to regions. Meshing, crucial for accurate simulation, required refinement to achieve desirable  $Y^+$  values and low energy residuals.

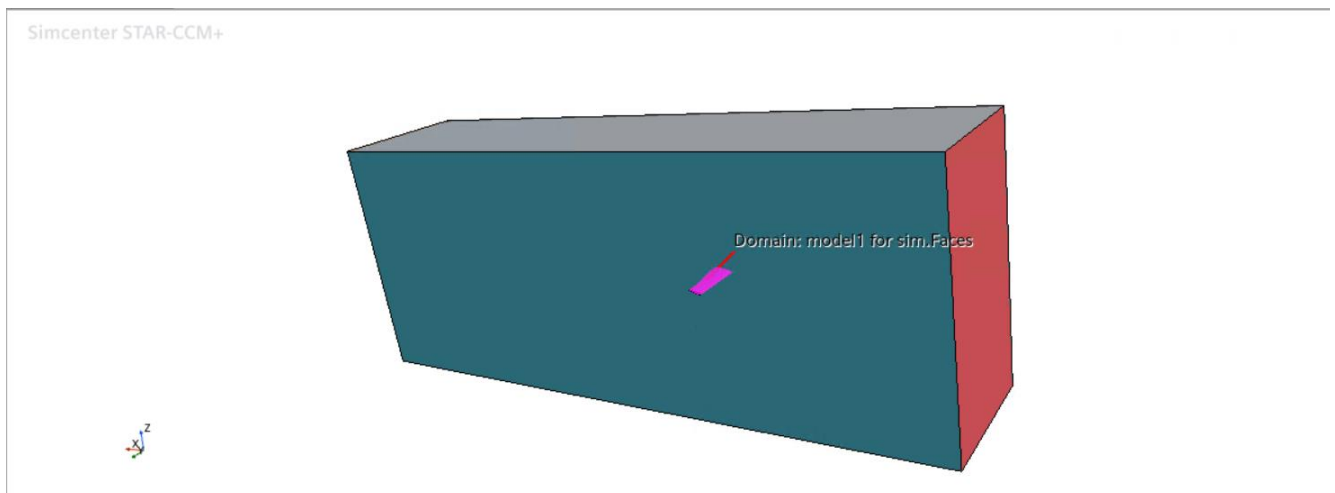


Figure 3: Domain

The simulation setup included steady flow, atmospheric pressure at sea level, 2D flow, turbulent viscous regime, segregated flow, and constant density. Mesh validation encompassed Mesh Quality Check, with specialized wake refinement downstream of the foil. Scene representations provided insightful depictions of velocity streamlines, wall  $Y^+$  function, pressure gradient, and velocity gradient across the wing surface, facilitating analysis.

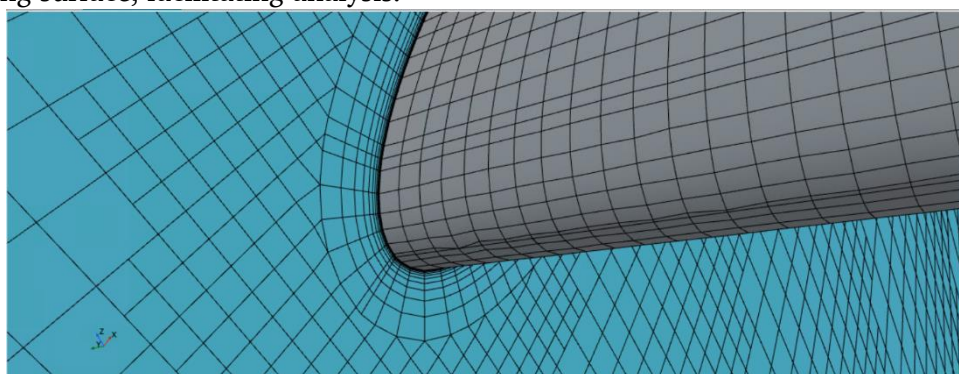


Figure 4 : Mesh at Leading Edge



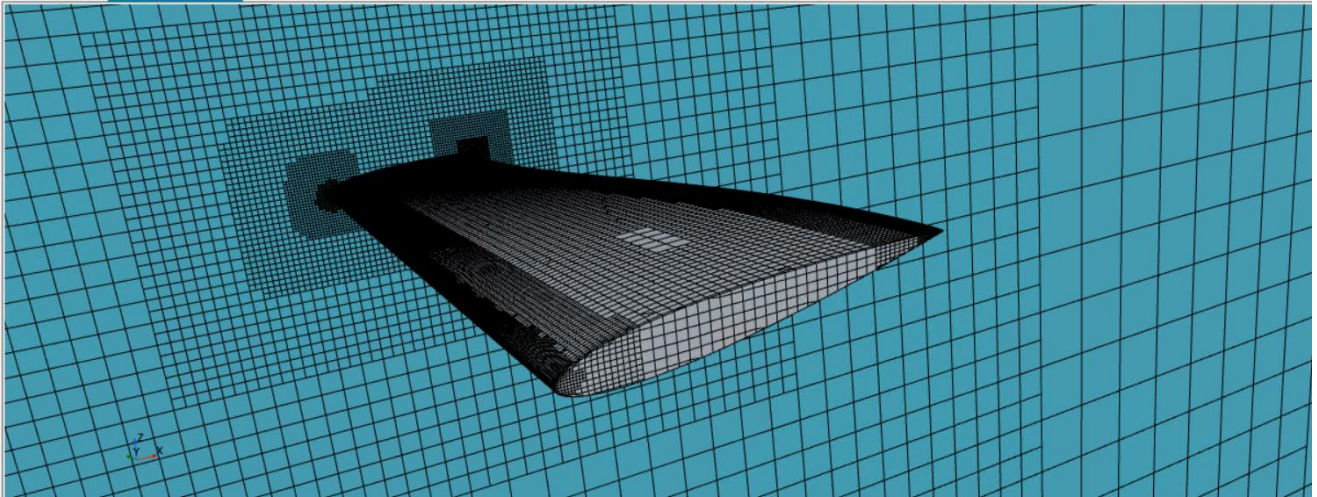


Figure 5: Mesh in trimetric view at wing

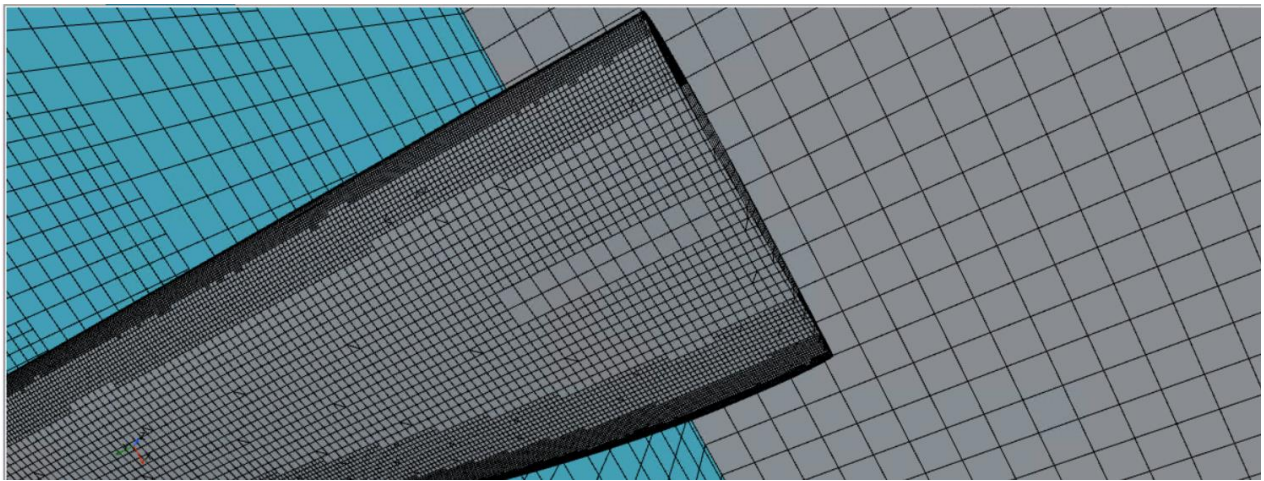


Figure 6: Mesh in Wing Profile

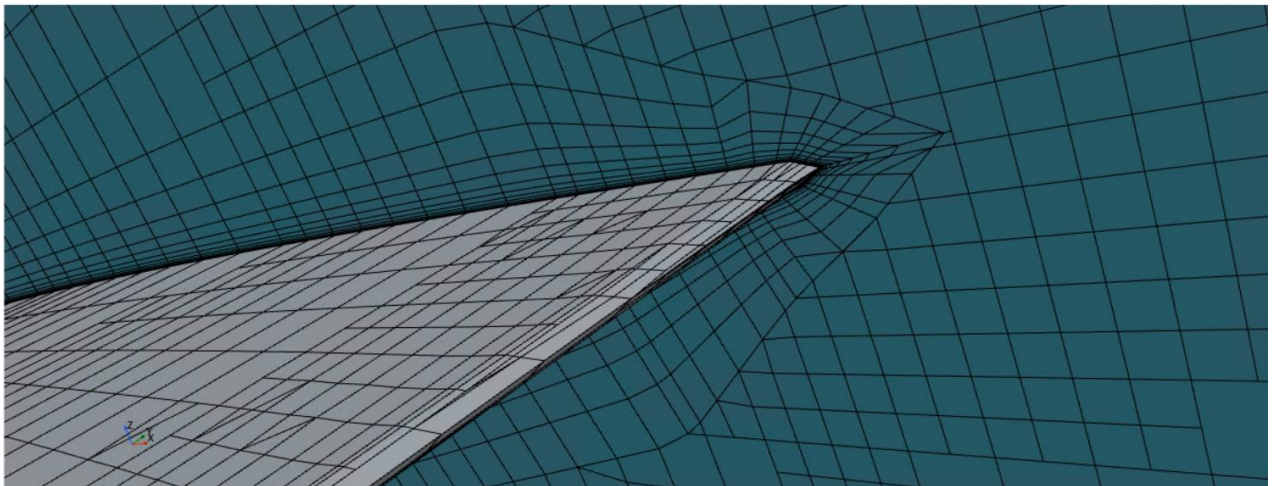


Figure 7: Mesh at Trailing Edge

The rigorous simulation and validation processes outlined in this report ensure accurate modeling and analysis of the spoiler's aerodynamic performance under various conditions, contributing to the advancement of automotive aerodynamics.

#### 4.3 Boundary Conditions

1. Inlet: Velocity Inlet Physics value= $67.06 \text{ m.s}^{-1}$

2. Outlet: Pressure Outlet Physics Value=Default
3. Top: Pressure Outlet Physics Value=Default
4. Bottom: Pressure Outlet Physics Value=Default
5. Side: Pressure Outlet Physics Value=Default
6. Root: Symmetry
7. Aerofoil Surface/Wing Surface: Wall Type Physics Value=Default

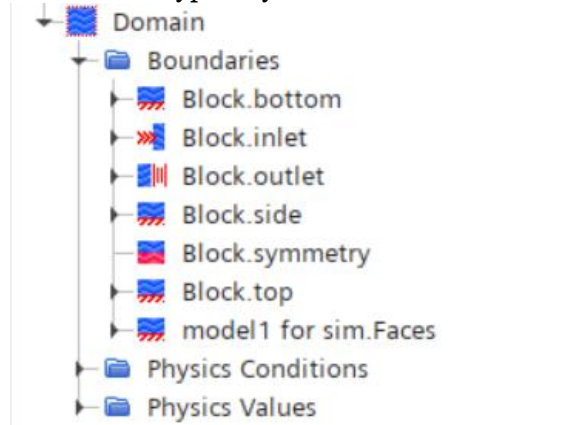


Figure 8: Boundary Conditions

#### 4.4 Physics Models

The selection of Star-CCM+ as the preferred CFD software stemmed from its superior accuracy compared to SolidWorks Flow Simulation. Star-CCM+ employs advanced numerical algorithms for solving the Navier-Stokes equations, enhancing precision in simulations of complex flows. Notably, the software utilizes the pressure-implicit with splitting of operators (PISO) method, known for its higher accuracy and robustness. This method employs a segregated solver approach, iteratively coupling velocity and pressure corrections to ensure mass conservation.

Star-CCM+ incorporates advanced solver technologies like the algebraic multigrid (AMG) method, which accelerates convergence of the pressure solver. Additionally, the software leverages a combination of K-epsilon and K-Omega methods to solve the Navier-Stokes equations more effectively. K-epsilon is preferred for analyses near object faces or walls, while K-Omega excels in solving Navier-Stokes equations in open volume flows, such as the wake behind the wing. The ability to utilize both methods significantly enhances accuracy.

Moreover, Star-CCM+ boasts a broader range of physics modeling capabilities compared to SolidWorks Flow Simulation. It offers finer meshing capabilities, allowing for local mesh refinement in critical areas like the leading edge of the wing while maintaining less refined meshing in less critical regions. The software's utilization of prism layers for boundary layer meshing further improves accuracy.

By refining the mesh in essential areas, Star-CCM+ optimizes accuracy while keeping simulation times manageable, a crucial factor in projects involving numerous simulations. The final simulation in this project was conducted in the environment of the following models: k-epsilon, Segregated Constant Temperature Flow, Steady, 3D, Turbulent Flow conditions, ensuring comprehensive analysis of the aerodynamic behavior of the spoiler.





Figure 9: Physics Models

## 5 Results and Discussion

The figure illustrates the downforce and drag characteristics of the wing at a velocity of 67.056 m/s. Our computations encompassed a range of speeds and angle of attacks from 0 to 83.25 degrees, considering the geometric limitations of the spoiler, which cannot tilt beyond 90 degrees. We focused our simulations between 10 and 15 degrees, as the critical angle falls within this range, as depicted on the graph.

At the critical angle, downforces experienced a sudden decrease while drag continued to increase. Flow separation at the critical angle leads to a significant decrease in downforce, while drag continues to rise. This data and plot informed our decisions regarding optimal angles for various driving scenarios. For instance, during simple acceleration, it is advantageous to maintain downforce while minimizing drag.

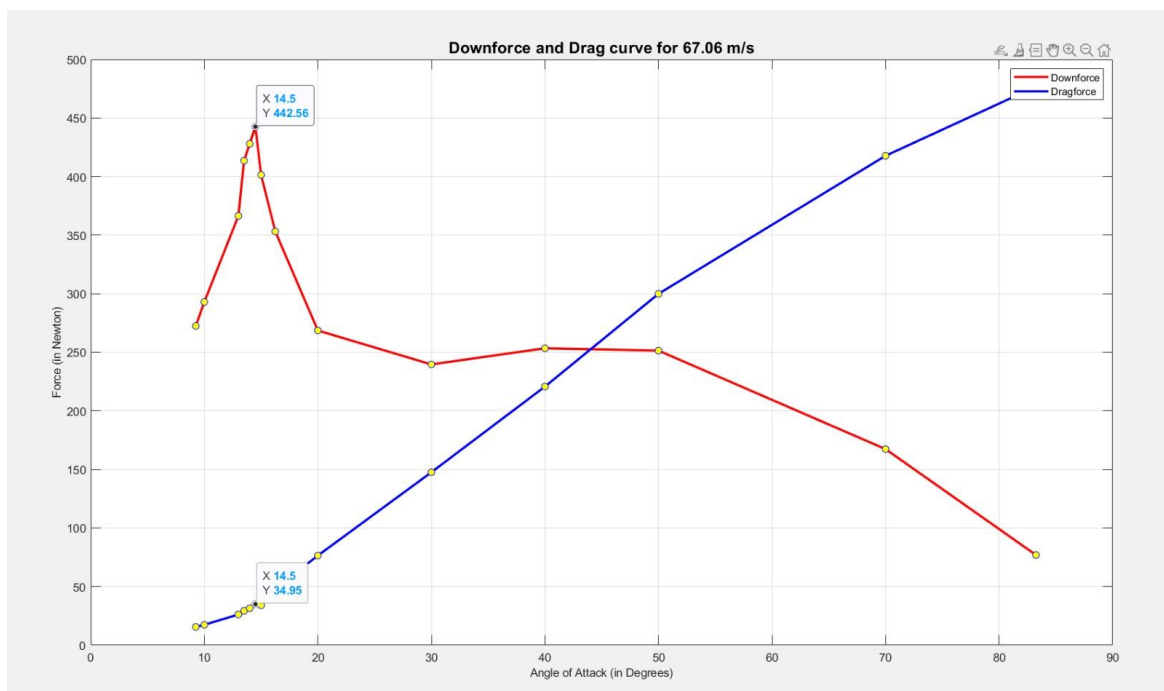


Figure 10 : Drag vs Downforce

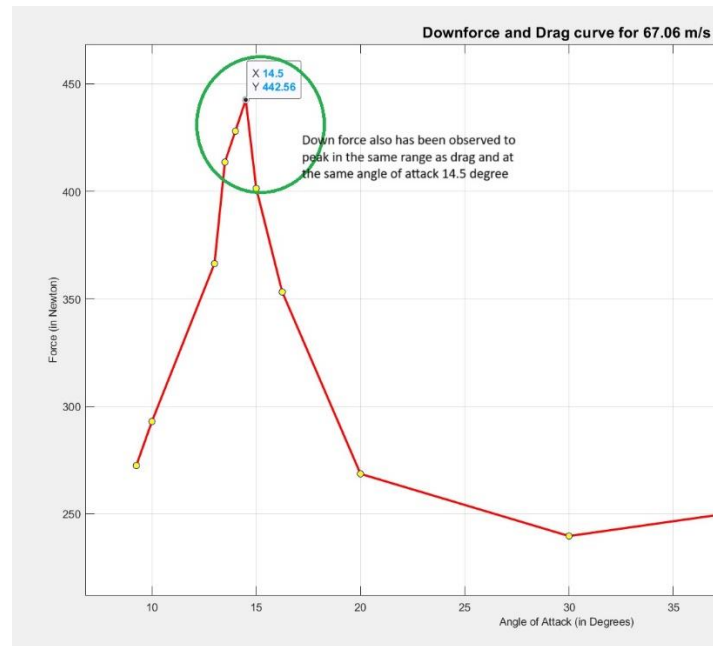


Figure 11: Downforce at 14.5 deg

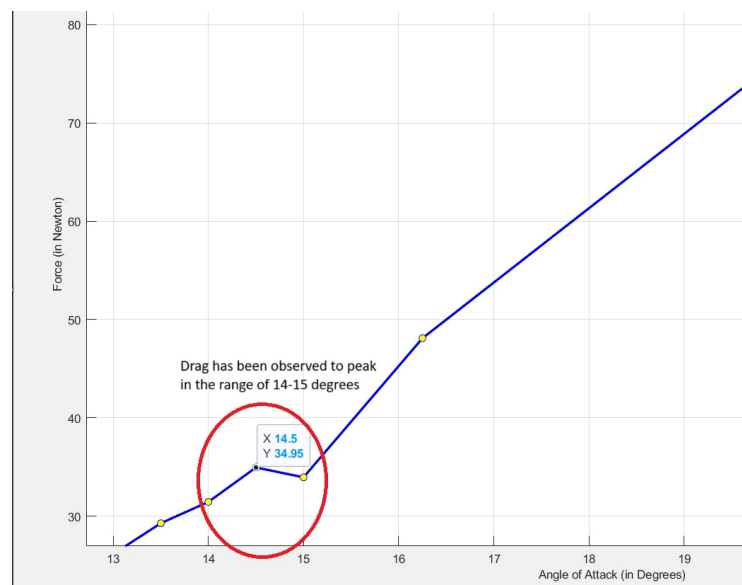


Figure 12: Drag at 14,5 deg

Therefore, in such situations, leaving the spoiler at 0 degrees provides downforce with minimal drag (refer to Figure 11). The drag spikes with downforce as it approaches the critical angle of attack, which proves beneficial when decelerating without stopping completely, as the downforce enhances wheel grip while the drag aids in deceleration. As anticipated, maximal drag occurs when the spoiler is fully tilted at 83.25 degrees, which can be advantageous in specific driving scenarios.

After conducting simulations across a wide range of angles in Star-CCM+, we determined that the most optimal angle for downforce was 14.5 degrees, yielding approximately 442.56 N of downforce. This angle is particularly suitable for cornering or driving through curves, where increased downforce prevents wheel slippage.

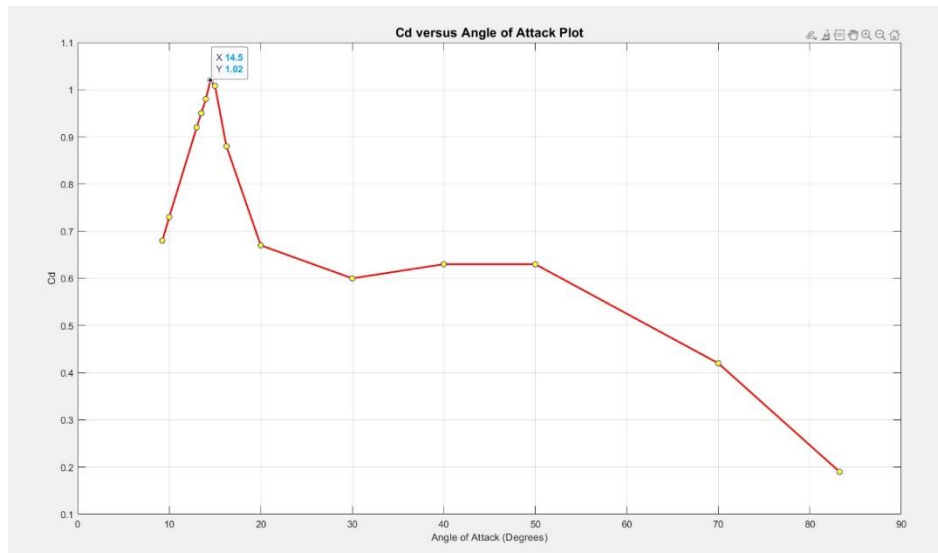


Figure 13:  $C_d$  vs Angle of Attack

As expected, both downforce and drag exhibit exponential increases with speed, given that downforce is proportional to the square of velocity.

As the vehicle attains higher velocities, the driver seeks enhanced downforce. Our analysis suggests that the spoiler angle should be set to 9.25 degrees, as observed from the intersection of the Drag and Downforce plots. Then the vehicle requires acceleration, minimal drag is desired, and downforce becomes irrelevant.

In such instances, the spoiler is ideally set at a zero angle of attack due to the symmetric nature of the NACA 0012 airfoil, where no lift or drag is generated, as indicated by our study and basic aerodynamic theory. During cornering maneuvers, maximum downforce is essential to prevent slippage. Tabular data indicates that an angle of attack of 15 degrees is capable of generating maximum downforce of 401.39N, facilitating better handling around corners.

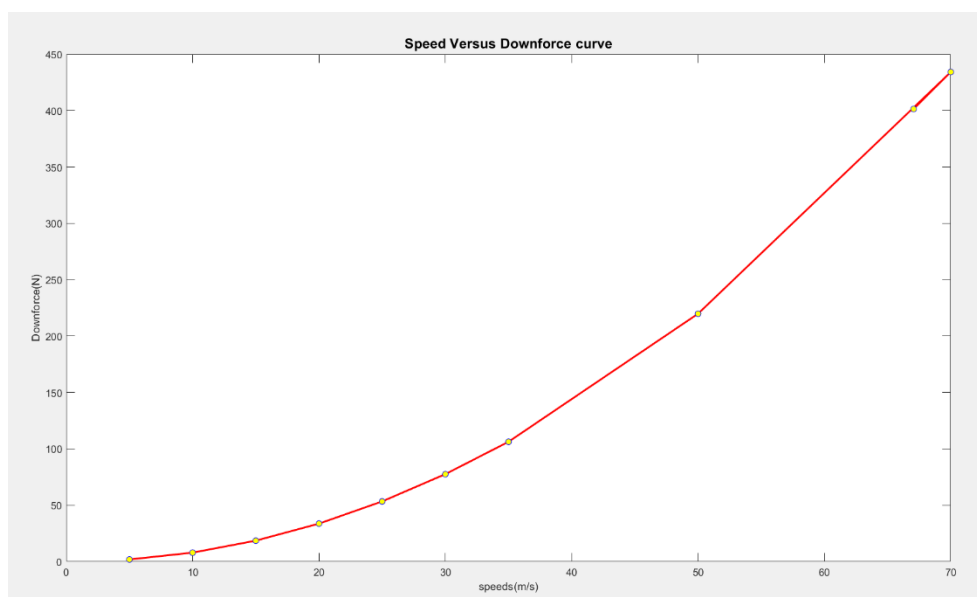


Figure 14 : Speed vs Downforce Plot

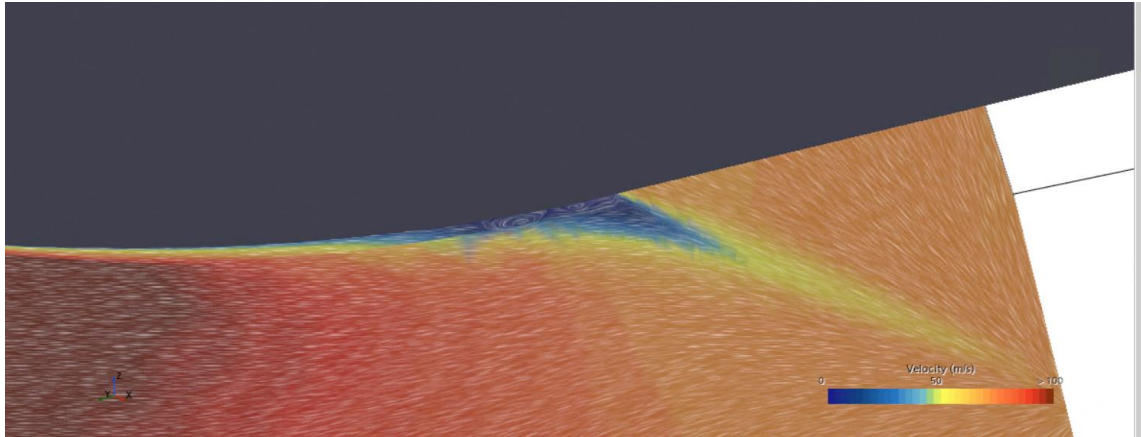


Figure 15 : Vortex Formation at 14.5 deg at bottom of Wing. This represents flow separation.

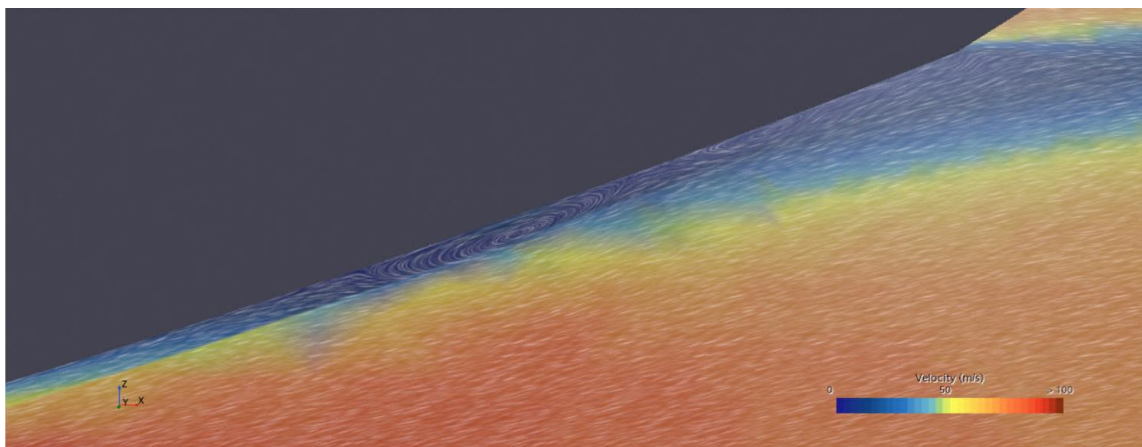


Figure 16 : Vortex Formation at 14.5 deg at bottom of Wing

Table 2 : Drag, Downforce and  $C_d$  readings

| Angle of Attack(Degrees) | Downforce(N) | Drag(N) | $C_D$ |
|--------------------------|--------------|---------|-------|
| 9.25                     | 272.44       | 15.51   | 0.68  |
| 10                       | 292.94       | 17.34   | 0.78  |
| 13                       | 366.386      | 26.15   | 0.92  |
| 13.5                     | 413.6        | 29.28   | 0.95  |
| 14                       | 428          | 31.45   | 0.98  |
| 14.5                     | 442.56       | 34.95   | 1.02  |
| 15                       | 401.39       | 33.95   | 1.008 |
| 16.25                    | 353.16       | 48.10   | 0.88  |
| 20                       | 268.58       | 76.44   | 0.67  |
| 30                       | 239.64       | 147.59  | 0.6   |
| 40                       | 253.41       | 220.64  | 0.63  |
| 50                       | 251.44       | 299.89  | 0.63  |
| 70                       | 167.34       | 417.75  | 0.42  |
| 83.25                    | 76.92        | 456.81  | 0.19  |

For more effective braking, higher levels of drag and downforce are required. Setting the spoiler at 15 degrees of angle of attack serves as the sweet spot for both drag and downforce, optimizing braking performance.

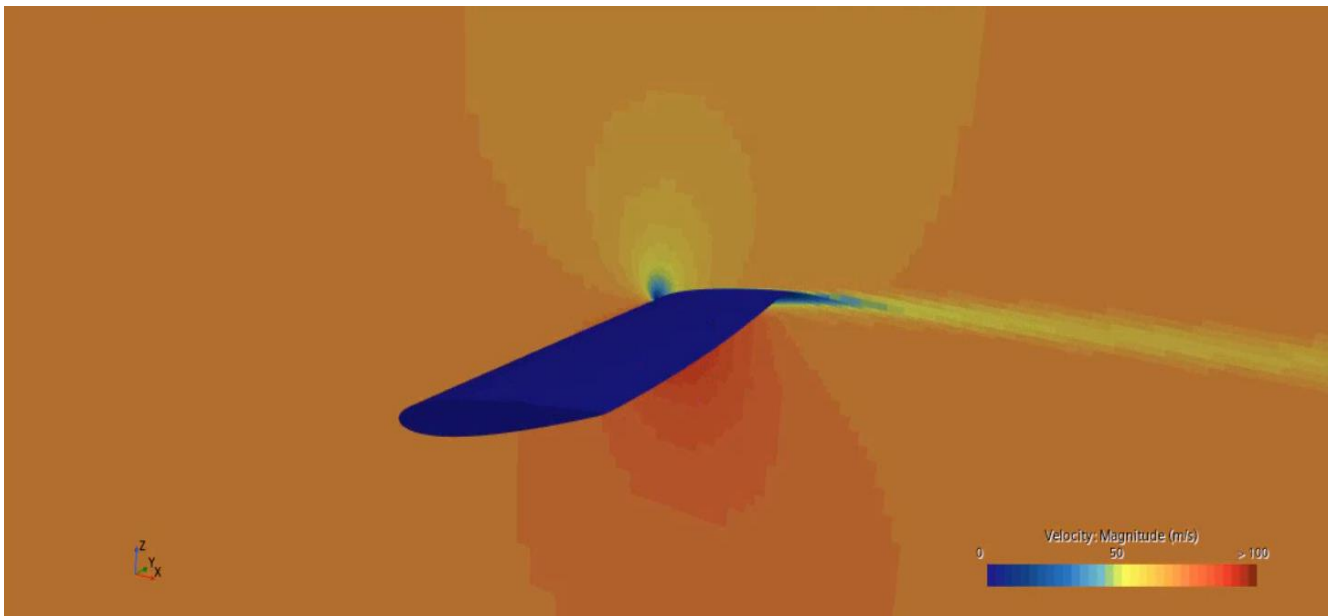


Figure 17 : Velocity Variation at Wing

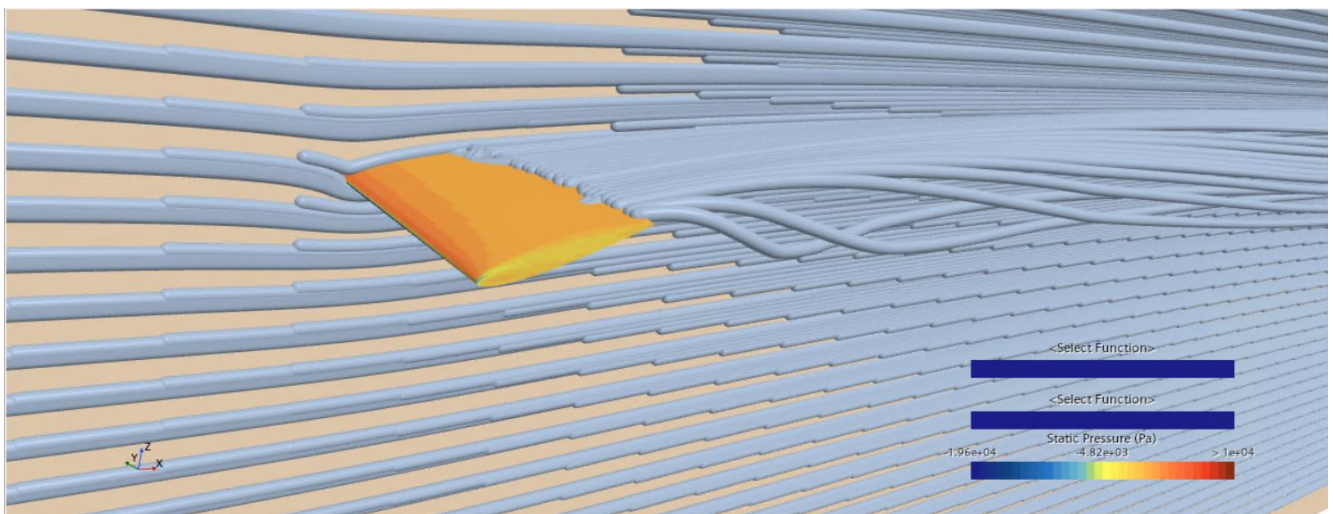


Figure 18 : Streamline at 14.5 deg representing flow field and flow separation



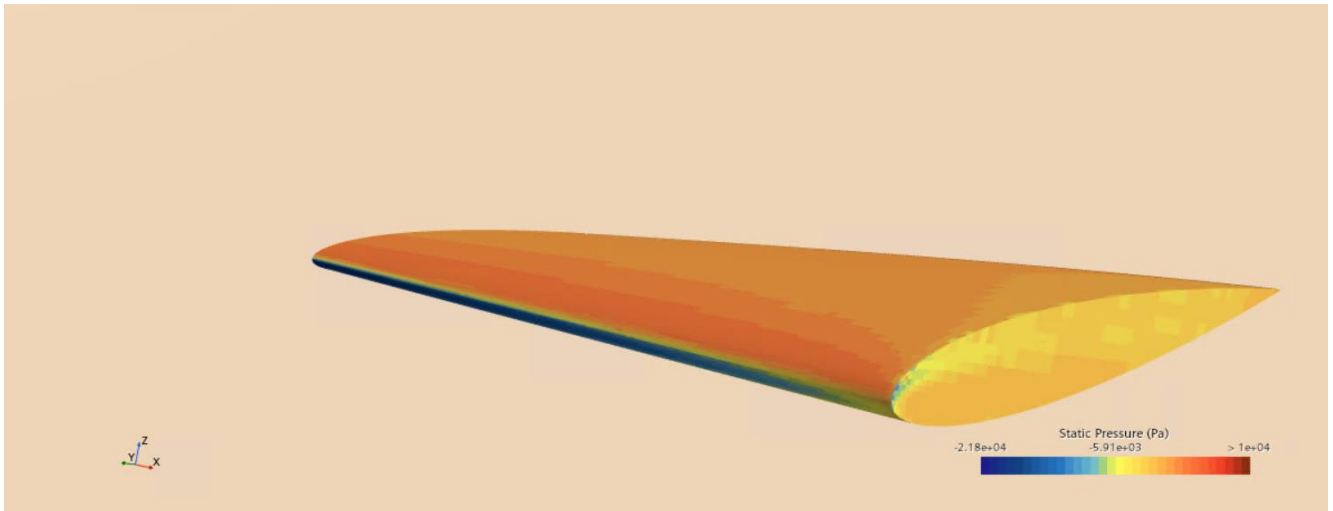


Figure 19 : Static Pressure variation in Wing Surface at 14.5 deg

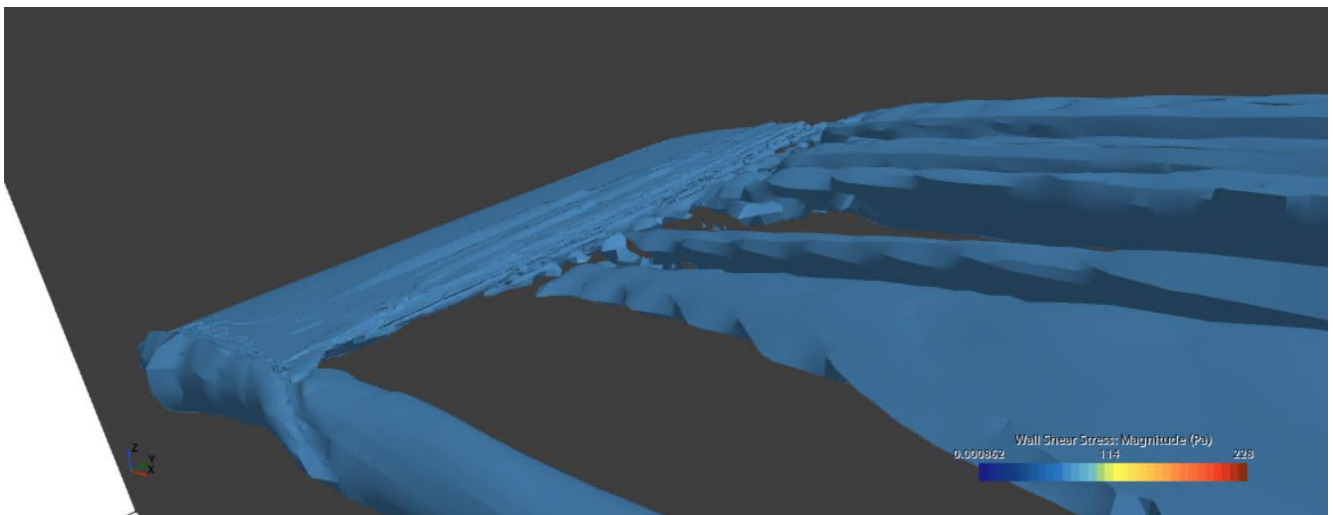


Figure 20 : Isosurface Representation for Vortices

### 5.1 Mesh Convergence Study

Based on our CAD model of the spoiler, we conducted a mesh dependence study using 33 different mesh sizes ranging from 1cm to 5000cm. The simulation was conducted at the maximum angle of attack to determine the maximum downforce. The cell count varied from 11,656 to 7,960,761, and the data was plotted against downforce using MATLAB, as depicted below.

From the graph, it is evident that the downforce stabilizes at higher cell counts. At a global mesh base size of 5cm, the observed downforce value is 442.56N, and the variation thereafter appears to be less than 0.2N, with the solution approximating to 42.56N at a mesh size of 1cm. However, the meshing time for the 1cm mesh size was considerably high at 270.35 minutes, compared to the 5-minute meshing time for the 5cm mesh size.

Therefore, the 5cm mesh size is deemed to be the optimal choice, as it balances computational efficiency and accuracy effectively, providing stable results with reasonable meshing times.

## 5.2 Mesh Quality Check and Result Validation

Measuring the validity and quality of mesh in STAR-CCM+ is done by examining the following parameters:

1. Cell skewness angle
2. Face validity metric
3. Cell quality metric
4. Volume change metric
5. Chevron quality indicator

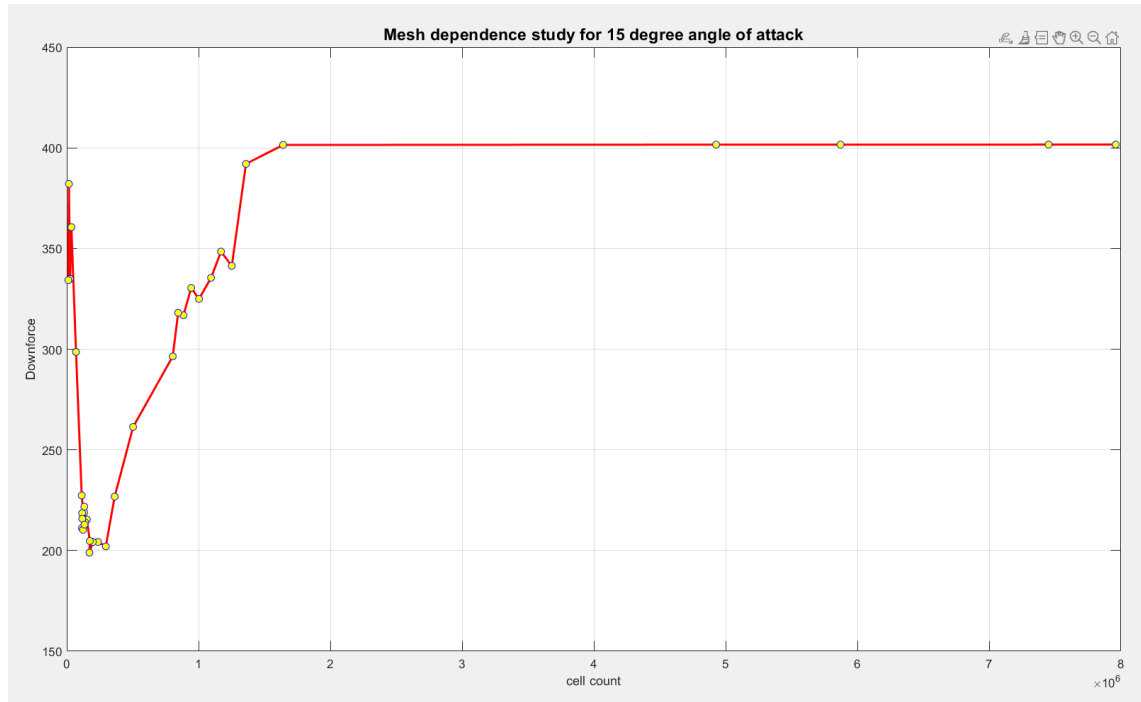


Figure 21: Downforce vs Cell Count in Millions

To evaluate the cell skewness of our mesh, we have made use of skewness angle test. Skewness angle less than 85 degrees indicates good quality mesh. Fortunately meshes in all of our domains defined was found to be less than 85 degrees indicating a good quality mesh in terms of skewness as the image below shows the plot of skewness of mesh. Also to further validate our results, we have done two more tests of face validity metric and cell quality metric. A face validity metric of 1 indicates good quality mesh and our face validity plot also indicates the same. The criterion for cell quality metric test is cell quality value of 1 indicates a perfect cell and a cell quality approaching zero (approximately  $<0.00001$ ) is considered to be a bad quality cell. For our case the lowest cell quality value was found in the range between 0.32 and 0.33.

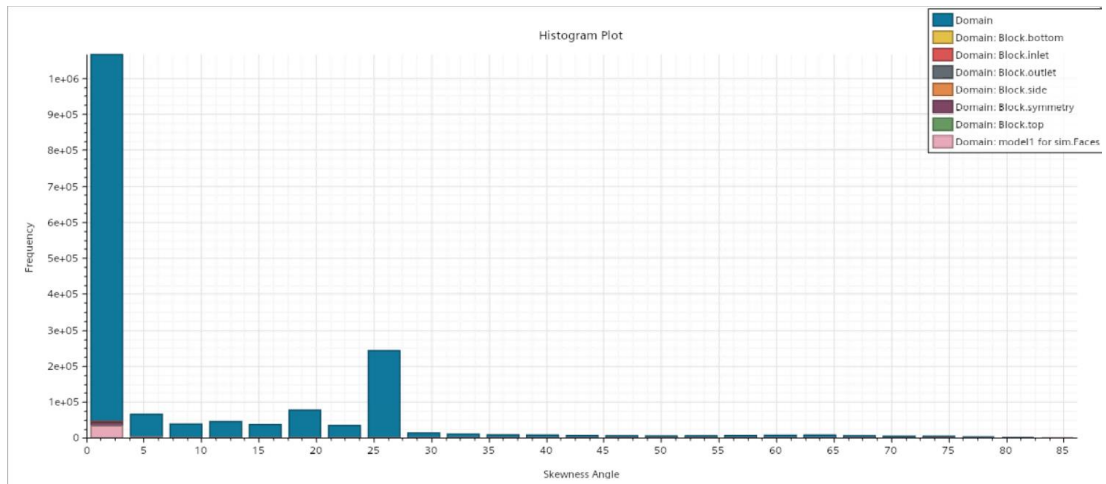


Figure 22 : Skewness Angle Test

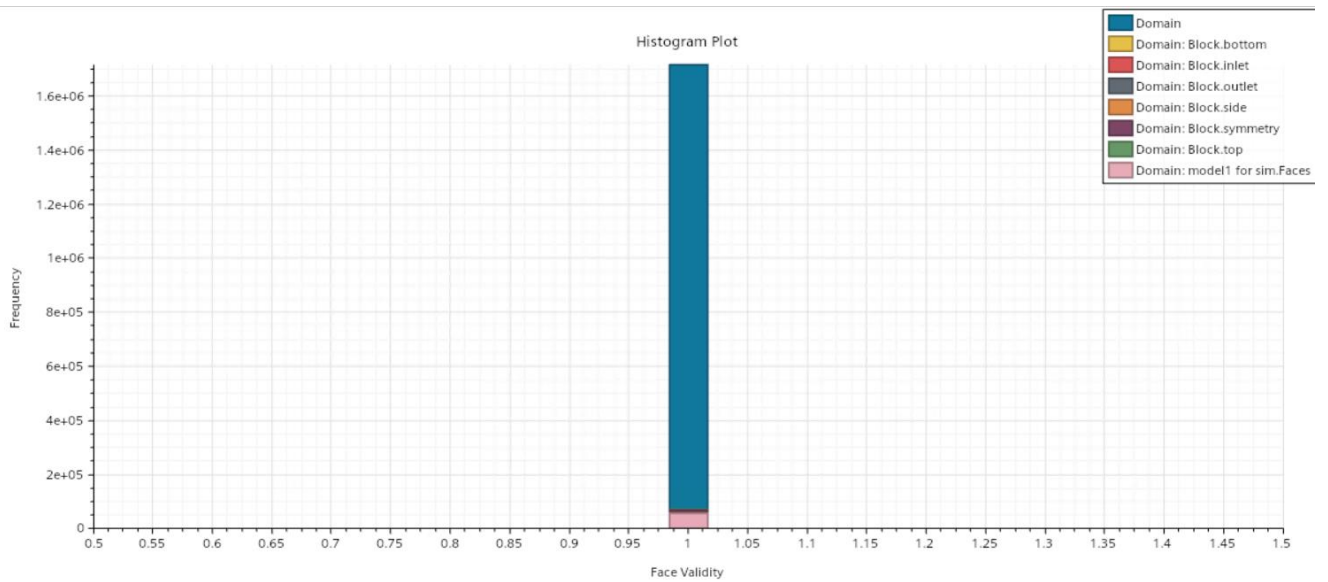


Figure 23 : Face Validity Test

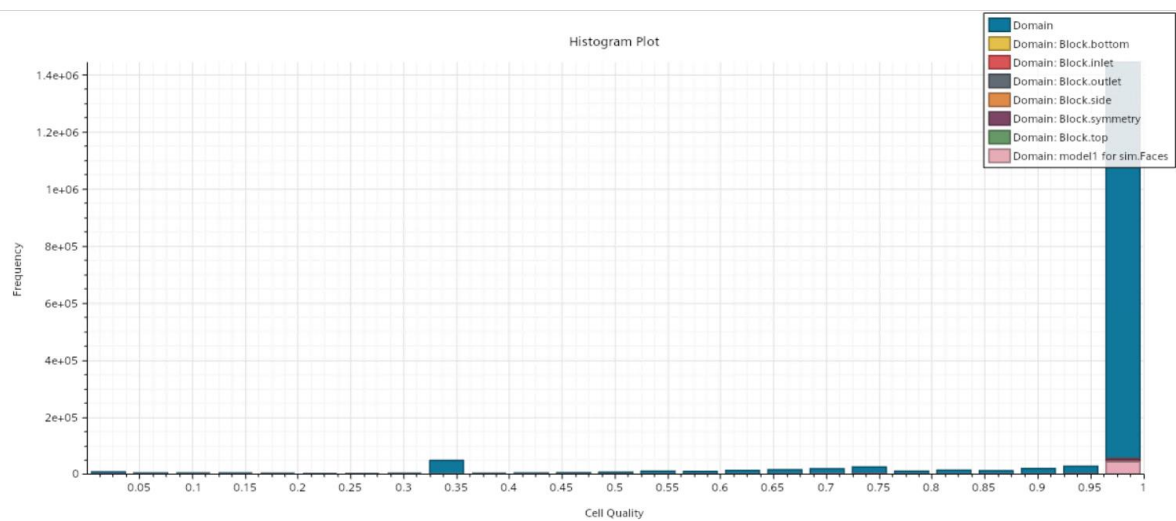


Figure 24: Cell Quality

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## 6. Conclusion

In the future, this product can be kept as an aftermarket product or integrated into sports cars by the manufacturers during production. In addition, this technology can also be used by racing teams to give themselves an upper hand against competitors. Aftermarket auto shops are another viable place where this technology can thrive. As the second aftermarket spoiler on the market, car enthusiasts can easily drop their vehicles off at a shop and have the active spoiler installed within the hour. Looking even further down the line, this technology can be modified and expanded upon to aid in vehicle efficiency and emissions for day to day commuters. To improve vehicle efficiency, the active aerodynamic features beyond just spoilers can be added to vehicles allowing the manipulation of air in order to maximize a vehicle's fuel efficiency or miles per charge. This technology can be extremely helpful for electric vehicles as their largest limitation is range. With an increase in efficiency, range can be increased allowing for electric vehicles to overcome their largest current pitfall.

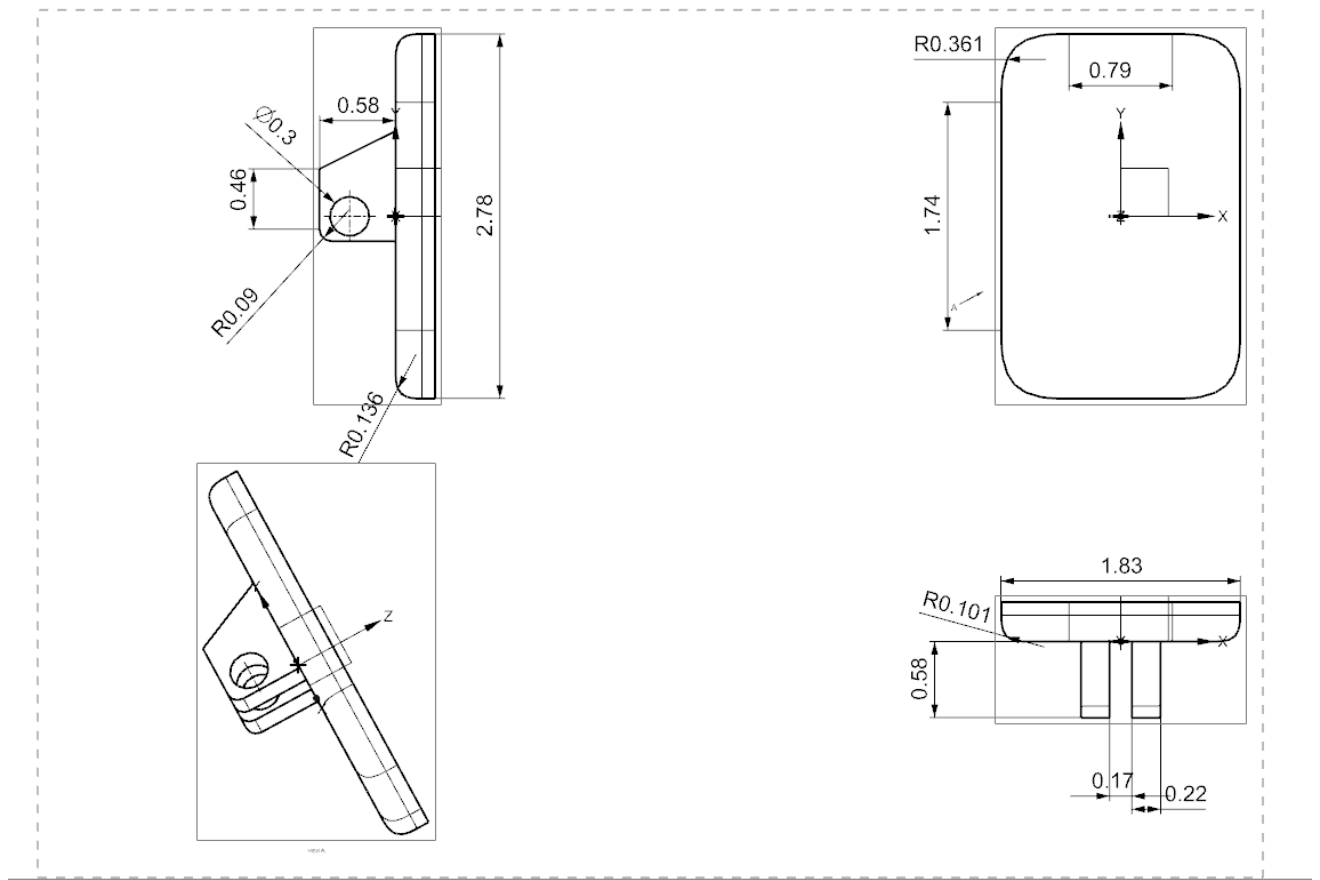
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## References

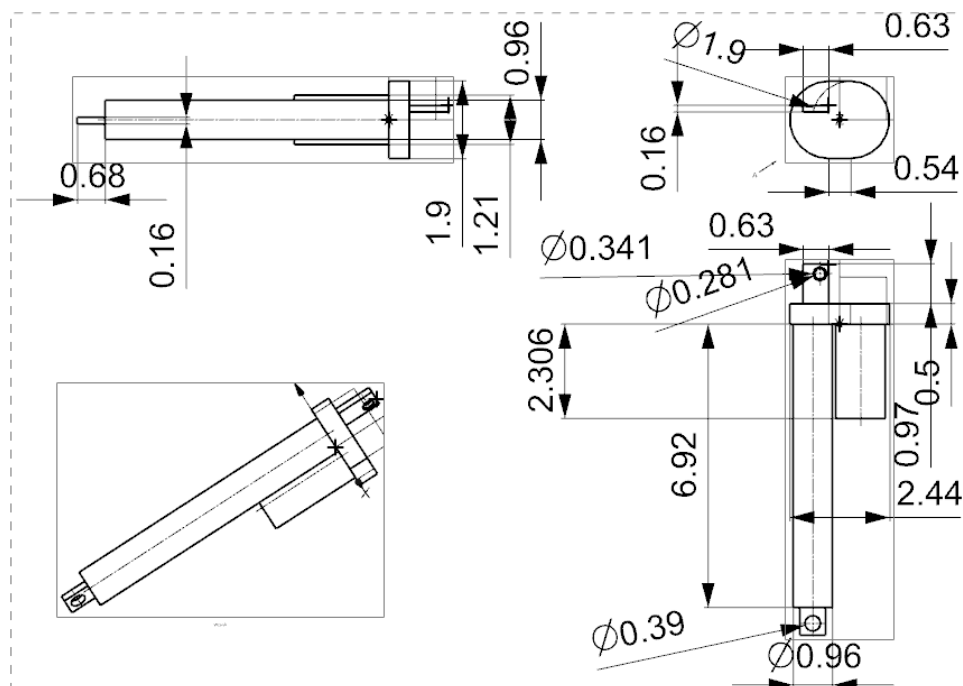
1. <https://youtu.be/v9OnNeYH4Ok?si=5NFn-bpKQhyxzAky>
2. <https://youtu.be/qAP2XNaxRxY?si=nvv9tq-BLZKPG-wn>
3. <https://youtu.be/pPKVjujZHKE?si=pKR3UFQKr6Krr2Nd>
4. [https://www.linkedin.com/posts/jousefmurad\\_machinelearning-synera-fem-activity-7152288907491962880-JPFH?utm\\_source=share&utm\\_medium=member\\_android](https://www.linkedin.com/posts/jousefmurad_machinelearning-synera-fem-activity-7152288907491962880-JPFH?utm_source=share&utm_medium=member_android)
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7. Lakshmanan, Balaji & Yadav, Surendra. (2020). Design And Analysis Of Rear Spoiler To Reduce Drag Coefficient Of A Sports Car.

## Appendix

### 1. Drawing Sheets of Components and Spoiler Assembly

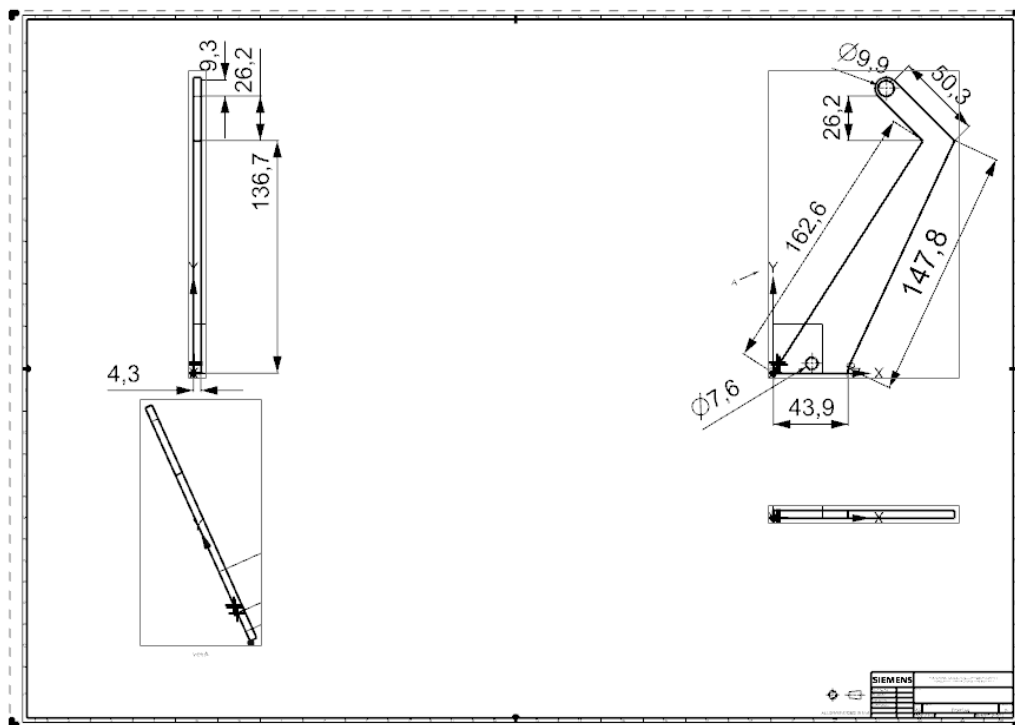


Appendix Figure 1: Back Mounting Plate

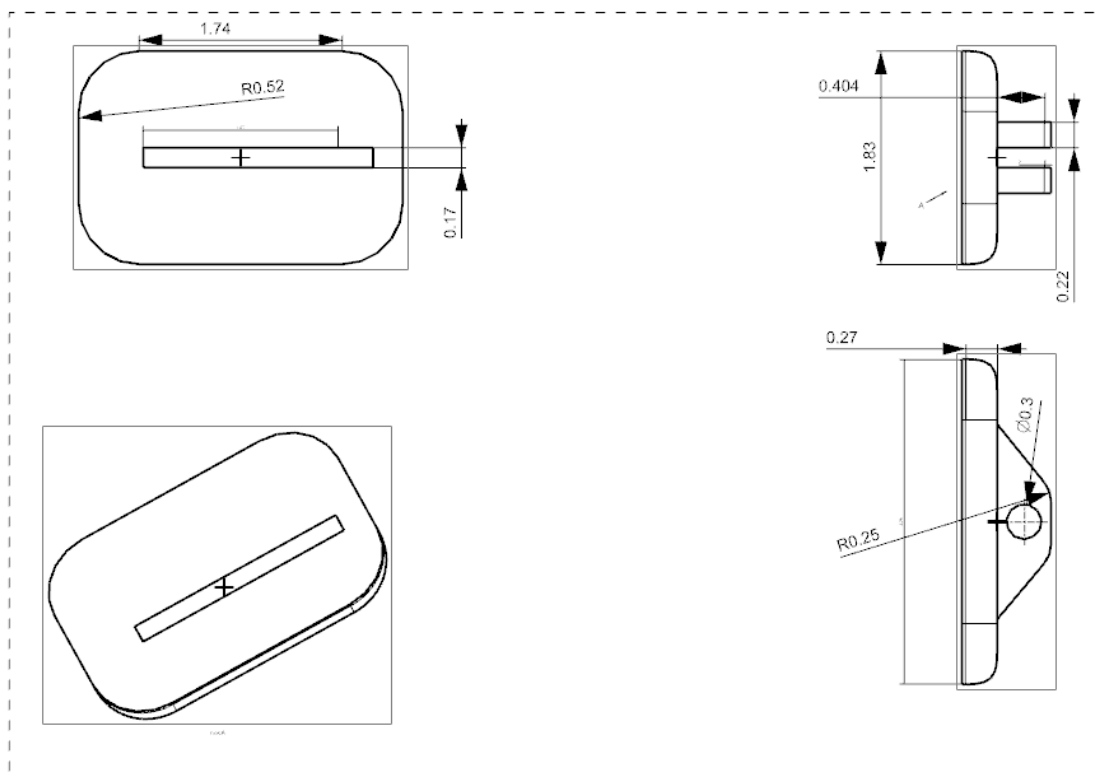


Appendix Figure 2: Back Leg

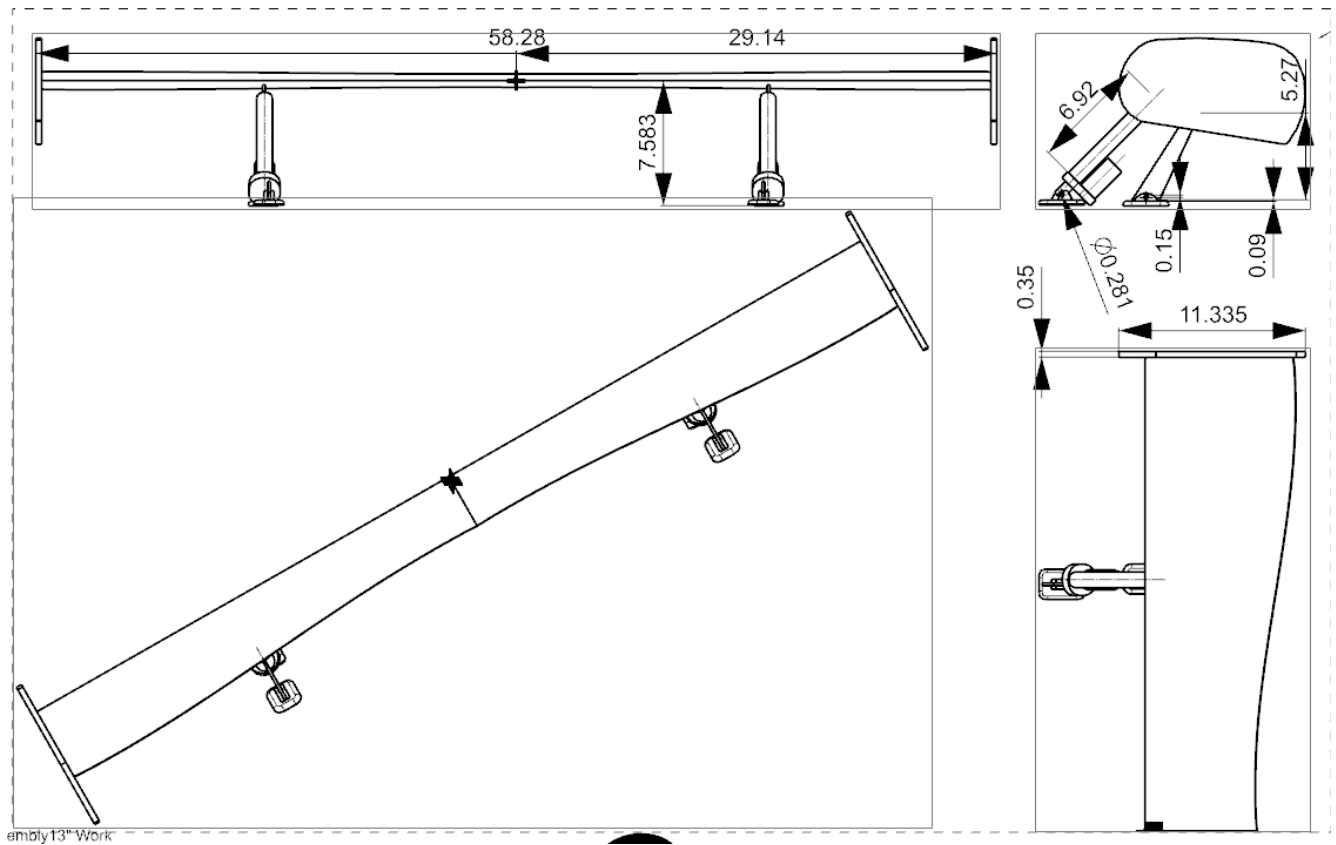




Appendix Figure 3: Front Leg



Appendix Figure 4: Front Mounting Plate



Appendix Figure 5 : Car Spoiler Assembly